

Faithfulness:

Field	Required for Faithfulness?
Input / Question	Usually provided to the evaluator/framework
RAG Context	✓ Yes
Actual Output	✓ Yes
Expected Output	✗ No

For example:

Input: “What is the refund period?”

Context: “Refunds can be requested within 30 days.”

Actual: “Refunds can be requested within 30 days and are processed within 2 days.”

Actual output has 2 claims; only 1 is supported by context:

$$\text{Faithfulness} = \text{Total Claims in Actual Output} / \text{Claims in Actual Output supported by Context}$$

Faithfulness=0.5

Case 1: Positive

The screenshot displays the Testleaf LLM Evaluation Framework interface. At the top, there are tabs for 'Single Evaluation', 'Batch Evaluation', 'Multi-Turn Evaluation', 'Golden Dataset', and 'Custom Metrics (B-Eval)'. The 'Single Evaluation' tab is active. Below the tabs, the 'Testleaf LLM Evaluation Framework' header is shown, followed by 'RAG & LLM Evaluation Platform'. The 'Active Framework' is set to 'DeepEval'. The 'Metrics' section includes 'All', 'Faithfulness', 'Answer relevancy', 'Contextual precision', 'Contextual recall', 'PI leakage', and 'Bias'. The 'Query/User Input' is 'What's your return window for electronics purchased on UrbanCart?'. The 'LLM Output/Actual Output From LLM' is 'You have 30 days from delivery to return electronics in their original packaging; opened electronics carry a 15% restocking fee unless the item is defective.'. The 'Retrieval From RAG Or From Document' is 'Electronics can be returned within 30 days of delivery in original packaging. || A 15% restocking fee applies to opened electronics unless defective.'. The 'Response' section shows '3,033 tokens', '1,208 in', '945 out', and '\$0.0000'. The 'FAITHFULNESS' metric has a 'SCORE' of '1.0000' and a 'VERDICT' of 'FAITHFUL'. The 'EXPLANATION' states 'The score is 1.00 because there are no contradictions, showing perfect faithfulness to the retrieval context.'. The 'CLAIM-LEVEL BREAKDOWN (1 CLAIM)' shows '1 supported', '0 contradicted', and '0 ambiguous'. The 'CLAIM' table has one row: 'You have 30 days from delivery to return electronics in their original packaging; opened electronics carry a 15% restocking fee unless the item is defective.' with a 'VERDICT' of 'YES' and a 'REASON' of '—'. The 'EVALUATION INPUT' section shows the 'Query' and 'Retrieved Context'.

Case 2:

Testleaf Evaluation Suite

Single EvaluationBatch EvaluationMulti-Turn EvaluationGolden DatasetCustom Metrics (G-Eval)

Testleaf LLM Evaluation Framework

RAG & LLM Evaluation Platform

Active Frameworks: DeepEval

Metrics

AI

Faithfulness

Answer relevancy

Contextual precision

Contextual recall

PI leakage

Bias

Hallucination

Query/User Input

Do UrbanCart Plus members get free returns on all items?

LLM Output/Actual Output From LLM

Yes, UrbanCart Plus members get completely free returns on every item, including furniture and large appliances.

Retrieval From RAG Or From Document

UrbanCart Plus members receive free return shipping on all items except furniture and large appliances. || Furniture and large appliance returns incur a \$40 pickup fee for Plus members.

Add Context

Click for Evaluation

Response

HTML ReportExcel Report

3,234 tokens1,232 in1,341 out\$0.0021

MetricFaithfulness

Score0.0000

VerdictNOT FAITHFUL

EXPLANATION

The score is 0.00 because the output claimed completely free returns on every item, contradicting the context that UrbanCart Plus members incur a \$40 pickup fee for furniture and large appliances.

CLAIM-LEVEL BREAKDOWN (1 CLAIM)

CLAIM

VERDICT

REASON

UrbanCart Plus members get completely free returns on every item, including furniture and large appliances.

NO

The context states UrbanCart Plus members do not get free returns for furniture and large appliances. They incur a \$40 pickup fee, which directly contradicts the claim of completely free returns on every item.

EVALUATION INPUT

Query: Do UrbanCart Plus members get free returns on all items?

Output: Yes, UrbanCart Plus members get completely free returns on every item, including furniture and large appliances.

Retrieved Context: UrbanCart Plus members receive free return shipping on all items except furniture and large appliances. || Furniture and large appliance returns incur a \$40 pickup fee even for Plus members.

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Case 3:

Testleaf Evaluation Suite

Single EvaluationBatch EvaluationMulti-Turn EvaluationGolden DatasetCustom Metrics (G-Eval)

Testleaf LLM Evaluation Framework

RAG & LLM Evaluation Platform

Active Frameworks: DeepEval

Metrics

AI

Faithfulness

Answer relevancy

Contextual precision

Contextual recall

PI leakage

Bias

Hallucination

Query/User Input

What happens if I overdraft my checking account?

LLM Output/Actual Output From LLM

You'll be charged a \$34 overdraft fee per transaction, and there's no cap on how many times this can happen in a day.

Retrieval From RAG Or From Document

Overdrafts incur a \$34 fee per transaction, capped at 3 fees per day. || Customers enrolled in Overdraft Protection instead have funds transferred from a linked savings account for a \$10 transfer fee.

Add Context

Click for Evaluation

Response

HTML ReportExcel Report

3,243 tokens1,232 in1,331 out\$0.0021

MetricFaithfulness

Score0.5000

VerdictPARTIAL

EXPLANATION

The score is 0.50 because the actual output incorrectly claims there is no fee cap, contradicting the context that limits fees to three per day.

CLAIM-LEVEL BREAKDOWN (2 CLAIMS)

CLAIM

VERDICT

REASON

You'll be charged a \$34 overdraft fee per transaction.

YES

There's no cap on how many times this can happen in a day.

NO

The context states the fee is capped at three fees per day, so the claim that there is no cap is directly contradicted.

EVALUATION INPUT

Query: What happens if I overdraft my checking account?

Output: You'll be charged a \$34 overdraft fee per transaction, and there's no cap on how many times this can happen in a day.

Retrieved Context: Overdrafts incur a \$34 fee per transaction, capped at 3 fees per day. || Customers enrolled in Overdraft Protection instead have funds transferred from a linked savings account for a \$10 transfer fee.

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Answer Relevancy

The question is

“Does the generated answer actually address the user's input/question?”

Unlike Faithfulness, RAG context is generally not the primary comparison.

Fields required:

Field	Required?	Purpose
Input / Question	✓	What the user asked
Actual Output	✓	What the LLM generated
RAG Context	✗	Not required for basic answer relevancy
Expected Output	✗	Not required for basic answer relevancy

Answer Relevancy = Similarity(Input, Actual Output)

Case 1:

The screenshot displays the Testleaf LLM Evaluation Framework interface. At the top, there's a navigation bar with options like 'Single Evaluation', 'Batch Evaluation', 'Multi-Turn Evaluation', 'Golden Dataset', and 'Custom Metrics (G-Eval)'. The main header reads 'Testleaf LLM Evaluation Framework' and 'RAG & LLM Evaluation Platform'. Below this, the 'Active Framework' is set to 'DeepEval'. The 'Metrics' section shows 'Answer relevancy' selected. The 'Query/User Input' is 'How do I apply a promo code at checkout on UrbanCart?'. The 'LLM Output/Actual Output From LLM' is 'Enter your promo code in the "Promo Code" field on the checkout page, just above the order total, then click Apply - the discount will reflect immediately in your order summary.' The 'Retrieval From RAG Or From Document' section shows 'Context 1' with an 'Add Context' button. A 'Click for Evaluation' button is present. The 'Response' section shows '2,174 tokens', '1,567 in', '607 out', and '\$0.2014'. The 'METRIC' is 'ANSWER RELEVANCY', the 'SCORE' is '1.0000', and the 'VERDICT' is 'RELEVANT'. The 'EXPLANATION' states: 'The score is 1.00 because the answer fully addressed how to apply a promo code at checkout on UrbanCart with no irrelevant content. Great job!'. The 'EVALUATION INPUT' section shows the 'Query' and 'Output'.

Testleaf Evaluation Suite

Single Evaluation Batch Evaluation Multi-Turn Evaluation Golden Dataset Custom Metrics (G-Eval)

Testleaf LLM Evaluation Framework

RAG & LLM Evaluation Platform

Active Framework: DeepEval

Metrics

☐ All ☐ Faithfulness ☒ Answer relevancy ☐ Contextual precision ☐ Contextual recall ☐ PII leakage ☐ Bias ☐ Hallucination

Query/User Input *

How do I apply a promo code at checkout on UrbanCart?

LLM Output/Actual Output From LLM *

Enter your promo code in the "Promo Code" field on the checkout page, just above the order total, then click Apply - the discount will reflect immediately in your order summary.

Retrieval From RAG Or From Document

Context 1

+ Add Context

Click for Evaluation

Response

HTML Report Excel Report

2,174 tokens 1,567 in 607 out \$0.2014

METRIC

ANSWER RELEVANCY

SCORE

1.0000

VERDICT

RELEVANT

EXPLANATION

The score is 1.00 because the answer fully addressed how to apply a promo code at checkout on UrbanCart with no irrelevant content. Great job!

EVALUATION INPUT

Query:

How do I apply a promo code at checkout on UrbanCart?

Output:

Enter your promo code in the "Promo Code" field on the checkout page, just above the order total, then click Apply - the discount will reflect immediately in your order summary.

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Case 2:

Testleaf Evaluation Suite

Single EvaluationBatch EvaluationMulti-Turn EvaluationGolden DatasetCustom Metrics (G-Eval)

Testleaf LLM Evaluation Framework

RAG & LLM Evaluation Platform

Active Framework: DeepEval

Metrics

☐ All

☐ Faithfulness

☒ Answer relevancy

☐ Contextual precision

☐ Contextual recall

☐ PI leakage

☐ Bias

☐ Hallucination

Query/User Input

How do I request access to the shared finance drive?

LLM Output/Actual Output From LLM

Vantage Corp's IT department supports over 5,000 employees globally and recently upgraded to a new ticketing system.

Retrieval From RAG Or From Document

Context 1

Add Context

Click for Evaluation

Response

HTML ReportExcel Report

2,308 tokens1,550 in1,722 out\$0.0015

METRIC

ANSWER RELEVANCY

SCORE

0.0000

VERDICT

NOT RELEVANT

EXPLANATION

The score is 0.00 because the output only described the IT department's employee count and a ticketing system upgrade, which are unrelated to how to request access to the finance drive.

EVALUATION INPUT

Query

How do I request access to the shared finance drive?

Outputs

Vantage Corp's IT department supports over 5,000 employees globally and recently upgraded to a new ticketing system.

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Case 3:

Testleaf Evaluation Suite

Single EvaluationBatch EvaluationMulti-Turn EvaluationGolden DatasetCustom Metrics (G-Eval)

Testleaf LLM Evaluation Framework

RAG & LLM Evaluation Platform

Active Framework: DeepEval

Metrics

☐ All

☐ Faithfulness

☒ Answer relevancy

☐ Contextual precision

☐ Contextual recall

☐ PI leakage

☐ Bias

☐ Hallucination

Query/User Input

How do I apply a promo code at checkout on UrbanCart?

LLM Output/Actual Output From LLM

UrbanCart offers free shipping on orders over \$50 and has a rewards program where you earn 2 points per dollar spent. Check out our current deals page for more savings!

Retrieval From RAG Or From Document

Context 1

Add Context

Click for Evaluation

Response

HTML ReportExcel Report

2,093 tokens1,620 in1,073 out\$0.0018

METRIC

ANSWER RELEVANCY

SCORE

0.0000

VERDICT

NOT RELEVANT

EXPLANATION

The score is 0.00 because the response only mentions free shipping eligibility, a rewards program, and checking the deals page—none of which explain how to apply a promo code at checkout, so there is no relevant content to warrant a higher score.

EVALUATION INPUT

Query

How do I apply a promo code at checkout on UrbanCart?

Outputs

UrbanCart offers free shipping on orders over \$50 and has a rewards program where you earn 2 points per dollar spent. Check out our current deals page for more savings!

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Contextual Precision

Out of the contexts RAG retrieved, did it put the useful contexts near the top?

Field	Required?	Why?
Input / Question	✓	What information are we trying to retrieve?
Retrieved RAG Context	✓	Chunks/documents returned by the retriever
Expected / Golden Output	✓*	Helps determine which retrieved contexts are relevant
Actual Output	✗	Not needed to evaluate retrieval precision

Case 1:

Testleaf LLM Evaluation Framework
RAG & LLM Evaluation Platform

Active Framework: **DeepEval**

Matrix: ☒ All ☐ Faithfulness ☐ Answer relevancy ☒ Contextual precision ☐ Contextual recall ☐ F1 Package ☐ Bias

Testbed: ☐ Evaluation

Query/Answer Input: What's the maximum weight for standard shipping on UrbanCart?

Expected Output: (Required For Contextual Precision And Contextual Recall Metrics)
Up to 50 lbs for standard shipping, heavier items need freight shipping.

Retrieved From RAG Or From Document: Standard shipping covers packages up to 50 lbs. Items over 50 lbs require freight shipping with a separate quote. UrbanCart was founded in 2011.

Click For Evaluation

Response: 1,872 tokens / 1,328 in / 1,958 out / 85/100%

Metric: **CONTEXTUAL PRECISION** SCORE: **1.0000** VERDICT: **HIGH PRECISION**

EXPLANATION
The score is 1.00 because the rule (rule 1) is directly relevant to the reason stated. "The test states "Standard shipping covers packages up to 50 lbs," and "Items over 50 lbs require freight shipping," directly matching the expected answer," and there are no irrelevant rules to lower the ranking.

EVALUATION INPUT
Query: What's the maximum weight for standard shipping on UrbanCart?
Retrieved context: Standard shipping covers packages up to 50 lbs. Items over 50 lbs require freight shipping with a separate quote. UrbanCart was founded in 2011.

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Case 2:

Testleaf LLM Evaluation Framework
RAG & LLM Evaluation Platform

Active Framework: **DeepEval**

Matrix: ☒ All ☐ Faithfulness ☐ Answer relevancy ☒ Contextual precision ☐ Contextual recall ☐ F1 Package ☐ Bias

Testbed: ☐ Evaluation

Query/Answer Input: As a user, I want to search for products and filter results by price range and category so that I can find relevant items quickly.

Expected Output: (Required For Contextual Precision And Contextual Recall Metrics)
Test cases should cover combined price/category filtering with no full page reload.

Retrieved From RAG Or From Document: The search index is rebuilt nightly at 2 AM. Our platform supports 10 languages. Filters can be combined price range + category and results update without a full page reload.

Click For Evaluation

Response: 1,872 tokens / 1,328 in / 1,958 out / 85/100%

Metric: **CONTEXTUAL PRECISION** SCORE: **0.3333** VERDICT: **LOW PRECISION**

EXPLANATION
The score is 0.33 because the only relevant rule (rule 1) - "Filters can be combined price range + category and results update without a full page reload" - directly matches the expected test case requirement. - "Filters can be combined price range + category and results update without a full page reload" - directly matches the expected test case requirement. - "The search index is rebuilt nightly at 2 AM" - does not address combined price/category filtering or page reload - and rule 2 reason - "Our platform supports 10 languages" - is unrelated to the filtering or page reload context. - The relevant result is ranked low, resulting in a low contextual precision.

EVALUATION INPUT
Query: As a user, I want to search for products and filter results by price range and category so that I can find relevant items quickly.
Retrieved context: The search index is rebuilt nightly at 2 AM. Our platform supports 10 languages. Filters can be combined price range + category and results update without a full page reload.

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Case 3:

Testleaf Evaluation Suite

Single EvaluationBatch EvaluationMulti-Turn EvaluationGolden DatasetCustom Metrics (2-Eval)

Testleaf LLM Evaluation Framework

RAG & LLM Evaluation Platform

Active Framework: DeepEval

Metrics *

All

Faithfulness

Answer relevancy

Contextual precision

Contextual recall

PI leakage

Bias

Hallucination

Query/User Input *

How do I add a dependent to my health plan?

Expected Output * (Required For Contextual Precision And Contextual Recall Metrics)

Add within 30 days of a qualifying life event via the benefits portal.

Retrieved From RAG Or From Document *

Harborview Health offers wellness programs including gym discounts.

Dependents can be added within 30 days of a qualifying life event (birth, marriage, adoption) via the benefits portal.

Harborview's member app has a 4.5-star rating.

Add Context

Click for Evaluation

Response

HTML ReportExcel Report

2,771 tokens1,480 in1,370 out\$8,009

METRIC

CONTEXTUAL PRECISION

SCORE

0.5000

VERDICT

PARTIAL PRECISION

EXPLANATION

The score is 0.50 because an irrelevant node appears before the relevant one (rank 1: "The text mentions "Harborview Health offers wellness programs including gym discounts" which does not address how to add a dependent.") and although the relevant node is correctly placed at rank 2 ("It states "Dependents can be added within 30 days of a qualifying life event (birth, marriage, adoption) via the benefits portal," directly matching the expected answer"), a later irrelevant node at rank 3 ("The mention of "Harborview's member app has a 4.5-star rating" is unrelated to adding dependents") prevents a higher score.

EVALUATION INPUT

Query:

How do I add a dependent to my health plan?

Retrieved Context:

Harborview Health offers wellness programs including gym discounts.
Dependents can be added within 30 days of a qualifying life event (birth, marriage, adoption) via the benefits portal.
Harborview's member app has a 4.5-star rating.

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Contextual Recall

Did RAG retrieve enough information to produce my expected/golden answer?

Field	Required?
Input / Question	✓
Expected / Golden Output	✓
RAG Context	✓
Actual Output	✗

Precision vs. Recall—the easiest distinction

Suppose your expected answer needs A, B, and C.

RAG retrieves:

A, B, X, Y, Z

Contextual recall asks:

Did I retrieve everything I needed?

You needed A, B, and C, but only got A and B.

Contextual precision asks:

Of what I retrieved, how much was useful and well-ranked?

You retrieved useful A and B but also irrelevant X, Y, and Z.

So precision suffers.

In short:

Recall = Did I retrieve ALL the needed information?
Precision = Did I retrieve/rank the RIGHT information?

Case 1:

The screenshot shows the Testleaf LLM Evaluation Framework interface. The top navigation bar includes links for Single Evaluation, Batch Evaluation, Multi-Turn Evaluation, Golden Dataset, and Custom Metrics (Q-Eval). The main header is "Testleaf LLM Evaluation Framework" with the subtitle "RAG & LLM Evaluation Platform". The active framework is "DeepEval".

Metrics: All, Faithfulness, Answer relevancy, Contextual precision, Contextual recall (selected), PI leakage, Bias, Hallucination.

Query/User Input: What are the steps to set up direct deposit with Meridian Bank?

Expected Output: (Required For Contextual Precision And Contextual Recall Metrics)
1) Download form from online banking 2) Give to employer payroll 3) Activates in 1-2 pay cycles.

Retrieval From RAG Or From Document:
Step 1: Log into online banking and download your Direct Deposit form.
Step 2: Provide the form to your employer's payroll department.
Step 3: Direct deposit typically activates within 1-2 pay cycles.

Click for Evaluation

Response: HTML Report, Excel Report
1,888 tokens | 1,209 in | 682 out | \$0.0012

Metric: CONTEXTUAL RECALL
Score: 1.0000
Verdict: HIGH RECALL

EXPLANATION:
The score is 1.00 because sentence 1 aligns with node 1, sentence 2 aligns with node 2, and sentence 3 aligns with node 3 in the retrieval context, covering the expected output perfectly.

EVALUATION INPUT
Query: What are the steps to set up direct deposit with Meridian Bank?
Retrieved Context:
Step 1: Log into online banking and download your Direct Deposit form.
Step 2: Provide the form to your employer's payroll department.
Step 3: Direct deposit typically activates within 1-2 pay cycles.

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Case 2:

The screenshot shows the Testleaf LLM Evaluation Framework interface. The top navigation bar includes links for Single Evaluation, Batch Evaluation, Multi-Turn Evaluation, Golden Dataset, and Custom Metrics (Q-Eval). The main header is "Testleaf LLM Evaluation Framework" with the subtitle "RAG & LLM Evaluation Platform". The active framework is "DeepEval".

Metrics: All, Faithfulness, Answer relevancy, Contextual precision, Contextual recall (selected), PI leakage, Bias, Hallucination.

Query/User Input: What are the steps to set up direct deposit with Meridian Bank?

Expected Output: (Required For Contextual Precision And Contextual Recall Metrics)
1) Download form from online banking 2) Give to employer payroll 3) Activates in 1-2 pay cycles.

Retrieval From RAG Or From Document:
Step 1: Log into online banking and download your Direct Deposit form.

Click for Evaluation

Response: HTML Report, Excel Report
1,833 tokens | 1,169 in | 664 out | \$0.0012

Metric: CONTEXTUAL RECALL
Score: 0.3333
Verdict: LOW RECALL

EXPLANATION:
The score is 0.33 because only sentence 1 of the expected output is supported by node 1 in the retrieval context, while sentences 2 and 3 have no backing evidence.

EVALUATION INPUT
Query: What are the steps to set up direct deposit with Meridian Bank?
Retrieved Context:
Step 1: Log into online banking and download your Direct Deposit form.

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Negative Scenarios:

Testleaf Evaluation Suite

Single Evaluation

Batch Evaluation

Multi-Turn Evaluation

Golden Dataset

Custom Metrics (G-Eval)

Testleaf LLM Evaluation Framework

RAG & LLM Evaluation Platform

Active Framework: DeepEval

Metrics

All

Faithfulness

Answer relevancy

Contextual precision

Contextual recall

Pii leakage

Bias

Hallucination

Query/User Input

What's your return window for electronics purchased on UrbanCart?

LLM Output/Actual Output From LLM

You have 30 days from delivery to return electronics in their original packaging; opened electronics carry a 15% restocking fee unless the item is defective.

Retrieval From RAG Or From Document

Electronics can be returned within 30 days of delivery in original packaging. ||| A 15% restocking fee applies to opened electronics unless defective.

+ Add Context

Click for Evaluation

Evaluation Failed

Failed to evaluate with DeepEval

Network Error

Response

Click "Evaluate" to see results

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